FACLON LABS POWER PLANT PRACTICE · INDUSTRIAL AI

Industrial AI for boiler, turbine & auxiliaries

From DCS data to closed-loop control: lower net heat rate, lower auxiliary power and fewer forced outages — proven on captive and utility plants across India.

3–5% Net heat rate reduction	**24** Power plant AI use cases	**98.4%** Validated RCA recommendation accuracy

FACLON LABS AT A GLANCE

Leaders in industrial data integration, visualization and AI-based optimization

5B+ data points daily	**200+** deployments	**60k+** assets connected	**50+** customers	**120+** employees

LEADERSHIP

Rishi Sharma
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B.Tech, IIT Bombay

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B.Tech, IIT Bombay

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INDIA OPERATIONS
B.Tech, IIT Bombay

POWER PLANT SME

Dr. T.G. Sundara Raman

50+ years in power plant operations, energy efficiency and thermal engineering

- Sector Expert, Bureau of Energy Efficiency (Ministry of Power)
- Consultant & Adviser — Energy Efficiency & Decarbonization
- B.Tech, M.Tech, Ph.D. (patent in heat transfer) and postdoctoral work, IIT Madras

STRATEGIC VALUE DRIVERS

Measurable impact on efficiency and reliability

3–5% NHR ↓
EFFICIENCY
Continuous AI-driven optimization of boiler–turbine cycle efficiency, directly lowering the cost of steam and power and cutting emissions.

Higher availability
ASSET RELIABILITY
Early anomaly detection and predictive diagnostics minimize unplanned downtime, extend equipment life and cut major overhaul costs.

SOLUTION ARCHITECTURE

Power plant digital twin

One platform, on-premise or cloud, from plant data sources to operator-facing outcomes.

DATA SOURCES
- DCS / SCADA — via OPC server, historian or data lake
- Lab data — coal, water
- Energy meters
- Design data — manuals, HMBD

→

DATA INTEGRATION
- Historian
- IT–OT integration / API
- Forms / ERP / LIMS
- File upload
- **Faclon AI Platform** — On-premise / cloud

→

OUTCOMES
- KPI benchmarking
- RCA & recommendations
- Optimal control strategy
- APC reduction
- Operator chatbot

Integration architecture for DCS, lab and energy-meter data is finalized during technical discussions with the plant team.

ENGAGEMENT APPROACH

From data exploration to plant-wide scale-up

Every engagement starts with the plant's own data and the people who run it.

1. **Data Exploration** — 1–2 WEEKS — Sample DCS, lab and design data reviewed offline. Tag coverage, data quality and loss profile screened against the use-case catalogue.
2. **Plant Visit** — 2–3 DAYS ON SITE — Walkdown of boiler, turbine and auxiliaries with our SME. Sessions with operations, maintenance, efficiency and C&I teams.
3. **Use Case Identification** — 1–2 WEEKS — Shortlist agreed on data availability, boiler type and value at stake. Baseline, KPIs and success criteria fixed in writing.
4. **Pilot** — 8–12 WEEKS — One or two use cases end to end: integration, model build, advisory in operators' hands, benefit measured against baseline.
5. **Scale-Up** — PHASED ROLLOUT — Remaining use cases and units on one platform. Advisory moves to closed loop where the pilot proves out.

Nothing is committed to the DCS before it is validated offline and accepted by operations.

PROOF

Customer cases

From 21 MW cogeneration to five-plant utility fleets.

SPB
21 MW cogeneration · 16 MW recovery boiler

Automated RCA Agents
AI agents on DCS and lab data that flag deviations across boiler, turbine, generator and auxiliaries and return root cause plus recommendation, built with a domain SME.
100+ RCAs generated daily · **98.4%** validated recommendation accuracy

Closed-loop Soot Blower Automation
AI-driven blowing across 29 soot blowers on a 16 MW recovery boiler — fewer wash cycles, less tube erosion, better flue-gas heat recovery.
20 TPD aux steam saved · **₹1 cr** annual savings

APC for Coal Feeding
Automated process control across 7 coal feeders of the AFBC boiler to cut process variability and hold bed stability.
7 feeders under APC · IN PROGRESS

JSW Energy
Five plants · Predictive maintenance

Vibration-based Predictive Maintenance
1,000+ vibration sensors deployed across five plants, cutting unplanned downtime, revenue loss and maintenance cost.
15+ critical failures caught early · **₹1.3 cr** savings over 2 yrs, finance-validated

Hindustan Zinc
Zawar · 91 MW captive plant

APC & NHR Reduction
Integrated with historian (DCS), EMS and lab data. Efficiency curves of all auxiliaries re-traced; run cycles scheduled to load, coal and season. NHR models handle IEX-driven load swings and steer operation toward HMBD reference ranges.
1 pp APC reduction target · IN PROGRESS

Grasim
175 MW captive · 5 boilers · 5 turbines

APC, Reliability & NHR Reduction
AI use cases to reduce net heat rate across a multi-unit captive plant.
5 + 5 boilers + turbines · IN PROGRESS

Tata Power
Large thermal plant · Prayagraj

Energy Management
Integrated with energy meters to monitor and dynamically benchmark APC equipment — boiler feed pumps, ID and FD fans.
Live APC benchmarking

USE CASE CATALOGUE

24 power plant AI use cases at a glance

Savings are indicative ("up to", per 100 MW) and are re-baselined against your plant's own data during use-case identification.

Process Data Analytics Monitoring, diagnostics and advisory · **Automated Process Control** Setpoint optimization, advisory or closed loop

01 · OVERALL EFFICIENCY			
01	SME Co-pilot	Process Data Analytics	₹5 cr
02	Digital Twin Simulation	Process Data Analytics	₹4 cr
02 · PROCESS CONTROL			
03	Multi-Feeder Coal Load Balancing + Bed Temperature Stabilization	Automated Process Control	₹5 cr
04	Air-Fuel Ratio Optimizer	Automated Process Control	₹4 cr
05	Attemperation Optimization	Automated Process Control	₹3 cr
06	Condenser–Cooling Tower Vacuum Optimizer	Automated Process Control	₹4 cr
07	Mill Fineness & PF Granulometry Optimization	Automated Process Control	₹3 cr
08	Multi-Coal Blend Advisory	Process Data Analytics	₹3 cr
09	Limestone Dosing & SO_2 Capture Optimization	Automated Process Control	₹2 cr
10	Furnace Draft & Windbox Pressure Balancing	Automated Process Control	₹2 cr
11	NOx Prediction & SCR/SNCR Reagent Optimization	Automated Process Control	₹2 cr
12	Bed Material Inventory & Cyclone Performance	Process Data Analytics	₹1.5 cr
03 · AUXILIARY POWER REDUCTION			
13	Aux Pumps + Fans Efficiency Retrace & Load Scheduling	Process Data Analytics	₹6 cr
14	Intelligent Cooling-Tower Fan Scheduling	Automated Process Control	₹2 cr
15	Mill Parameter Optimization	Automated Process Control	₹2 cr
16	Optimal ESP TR-Set Setpoint Control	Automated Process Control	₹2 cr
04 · ASSET RELIABILITY			
17	Vibration-Based Condition Monitoring	Process Data Analytics	₹5 cr
18	Intelligent Soot Blowing Scheduler	Automated Process Control	₹5 cr
19	Heat Exchange Index & Section-Level Fouling (incl. APH)	Process Data Analytics	₹3 cr
20	HP & LP Heater Performance Monitoring	Process Data Analytics	₹3 cr
21	ESP Health Monitoring	Process Data Analytics	₹2 cr
22	Generator Winding + Air Cooler Health	Process Data Analytics	₹2 cr
23	Corrosion & Scaling Surveillance	Process Data Analytics	₹2 cr
24	Condensate Contamination & Condenser Tube Leak Detection	Process Data Analytics	₹2 cr

01 Overall Efficiency

Plant-wide view of heat rate and its losses · 2 use cases

01 SME Co-pilot
Process Data Analytics · All boilers — **₹5 cr** up to, per 100 MW

SEES Benchmarks canonical KPIs live — net/gross heat rate, boiler and turbine efficiency, APC, condenser vacuum, excess O_2/CO, emissions margins, reliability — against HMBD design, shift target and golden band.

FINDS On deviation, cross-references symptoms across boiler, steam cycle, turbine, auxiliaries and emissions to rank probable root causes with confidence scores and cost attribution.

ACTS Answers operator queries through a chatbot and issues ranked corrective actions — e.g. *"cut excess O_2 from 4.2 to 3.5%", "address APH radial seal leakage", "redistribute BFP load to BFP #2".*

PROBLEMS TODAY KPI reporting fragmented across dashboards with no unified view · Operators don't know which lever to pull when a KPI drifts · Diagnostic know-how is tribal and retires with senior operators · Boiler and turbine teams troubleshoot in silos

02 Digital Twin Simulation
Process Data Analytics · All boilers — **₹4 cr** up to, per 100 MW

SEES First-principles heat and mass balance of the steam cycle, built from HMBD and OEM design data and continuously calibrated to live DCS and lab data.

FINDS Splits the NHR gap versus design into named losses — condenser vacuum, dry flue gas, unburnt carbon, APH leakage, spray, heater performance, APC — each in kcal/kWh and ₹/year.

ACTS Simulates what-ifs before touching the plant: coal blends, IEX-driven load ramps, an asset out of service, extraction changes. Sweeps setpoints for the feasible optimum at each load and validates closed-loop strategies offline before any setpoint reaches the DCS.

PROBLEMS TODAY Heat rate gap known only as a single number · Golden bands set from historical best bake in existing inefficiency · No safe way to test a control strategy before it touches the DCS

02 Process Control

Closed-loop and advisory control of combustion, steam and emissions · 10 use cases

03 Multi-Feeder Coal Load Balancing + Bed Temperature Stabilization
Automated Process Control · CFBC + AFBC — **₹5 cr** up to, per 100 MW

SEES Analyses bed temperature distribution and pressure drop, individual feeder rates, secondary air pattern, outlet steam stability and live coal quality.

FINDS Detects bed swings and feeder-level firing imbalance — the coupled bed-and-feeder problem that operator control treats as separate loops.

ACTS Pushes closed-loop bias setpoints to feeders and secondary air dampers, holding bed temperature tight and preventing sintering and defluidization. A stable bed steadies outlet steam temperature and cuts reliance on attemperation.

PROBLEMS TODAY Bed temperature swings drive efficiency loss and sintering risk · Multi-feeder coordination is manual and skill-dependent · Coal quality variability destabilizes the bed · Sintering and defluidization cause forced outages

04 Air-Fuel Ratio Optimizer
Automated Process Control · All boilers — **₹4 cr** up to, per 100 MW

SEES Analyses flue gas O_2 at economizer outlet, CO, NOx, FEGT, live coal quality and load.

FINDS Predicts the safe minimum excess O_2 setpoint for current coal and load — the efficiency that safety-margin setpoints leave on the table.

ACTS Pushes coordinated setpoints to fan demand, dampers and feeder rates, holding optimal O_2 with live CO-breakthrough protection.

PROBLEMS TODAY Plants run safety-margin excess O_2 · CO/O_2 trade-off left to operator judgment · Weak multivariate coordination across PA/FD fans, dampers and feeders · Coal quality changes cause excess-air swings

05 Attemperation Optimization
Automated Process Control · All (RH spray lever on reheat units) — **₹3 cr** up to, per 100 MW

SEES Analyses main steam and hot reheat temperature, SH/RH spray flow and valve position, degrees of superheat, section-wise SH/RH absorption, burner tilt (PF) or bed temperature (FBC), excess O_2 and soot-blowing state.

FINDS Separates spray genuinely needed for tube-metal protection from spray masking correctable upstream imbalance — fouling profile, air distribution, burner tilt, mill combination.

ACTS Corrects upstream levers first and trims spray only for residual control — cutting spray flow (especially costlier RH spray) and thermal fatigue at desuperheaters and outlet headers.

PROBLEMS TODAY Spray used as the primary steam temperature controller · RH spray's far larger heat-rate penalty is invisible to operators · Steam held below design to preserve spray margin · Attemperation masks upstream fouling and mill imbalance

06 Condenser–Cooling Tower Vacuum Optimizer
Automated Process Control · All boilers — **₹4 cr** up to, per 100 MW

SEES Analyses turbine exhaust enthalpy and back pressure, condenser TTD, CW supply/return temperatures, ambient wet bulb, load and air ingress.

FINDS Predicts the optimal trade-off between vacuum gain and the aux-power penalty of CT fans, CW pumps and vacuum pumps as ambient and load shift.

ACTS Advises CT fan cells ON, CW pump combination, header pressure trim, per-cell VFD speed and vacuum-pump duty cycle.

PROBLEMS TODAY Vacuum below design is a direct output loss · CW and CT operation static regardless of ambient/load · Air ingress hidden until significant · All CT cells run in mild weather

07 Mill Fineness & PF Granulometry Optimization
Automated Process Control · PF + SC + USC — **₹3 cr** up to, per 100 MW

SEES Analyses PF fineness, classifier setpoint, mill DP, outlet temperature, live coal particle size from computer vision and inferred grindability.

FINDS Predicts mill power versus fineness target per coal type, including the LOI penalty from coarse particles.

ACTS Advises classifier speed and mill DP to hold target fineness at minimum specific energy, each mill tuned to its coal source.

PROBLEMS TODAY Coarse PF raises LOI and unburnt carbon loss · Classifier settings rarely re-tuned · Over-fineness wastes mill power · Grindability variability ignored

08 Multi-Coal Blend Advisory
Process Data Analytics · Multi-source coal — **₹3 cr** up to, per 100 MW

SEES Analyses stockpile inventory by source, cost economics, coal quality, boiler behaviour per source and forward dispatch.

FINDS Predicts combustion, slagging, emissions, mill capacity and cost per kcal delivered for each candidate blend.

ACTS Issues a daily, explainable blend recommendation for joint operations and procurement — advisory, not auto-execute.

PROBLEMS TODAY Multiple sources with widely varying quality and cost · Blending decided weekly, not daily · Operability constraints learned by experience, not modeled · Cheap coals can cause expensive operating problems

09 Limestone Dosing & SO_2 Capture Optimization
Automated Process Control · CFBC only — **₹2 cr** up to, per 100 MW

SEES Analyses live stack SO_2, coal sulphur, bed temperature and limestone feeder rate against in-bed sulfation chemistry.

FINDS Predicts SO_2 capture efficiency across Ca:S ratios, including bed-temperature effects on kinetics.

ACTS Advises feeder setpoint to hold stack SO_2 in band at minimum Ca:S — cutting limestone use and improving fly-ash quality for cement sale.

PROBLEMS TODAY Limestone overdosed against SO_2 spikes · Compliance margins tightening under new norms · Coal sulphur varies cargo to cargo · Excess limestone hurts fly-ash sale value

10 Furnace Draft & Windbox Pressure Balancing
Automated Process Control · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses furnace draft at multiple points, windbox pressure across burner elevations (PF) or distributor-plate DP across bed sections (CFBC/AFBC), fan currents and damper positions.

FINDS Detects draft instability and windbox bias causing uneven combustion or non-uniform fluidization.

ACTS Provides coordinated fan-demand and damper trim for stable draft and uniform windbox pressure, with anticipatory adjustment on load ramps.

PROBLEMS TODAY Draft swings cause inefficient combustion and emission spikes · Windbox imbalance across burners · Damper coordination driven by habit · Slow load-following from manual fan adjustment

11 NOx Prediction & SCR/SNCR Reagent Optimization
Automated Process Control · SCR/SNCR equipped — **₹2 cr** up to, per 100 MW

SEES Models NOx formation upstream of the stack analyzer from burner conditions, FEGT, excess O_2, combustion staging and coal nitrogen.

FINDS Predicts NOx before the analyzer registers it, enabling anticipatory rather than reactive dosing.

ACTS Advises injection rate and stoichiometric ratio to hold compliance at lower reagent use, limiting ammonia slip and extending catalyst life.

PROBLEMS TODAY NOx limits tightening · Reagent dosed conservatively · Ammonia slip degrades expensive SCR catalyst · Manual reagent control is reactive

12 Bed Material Inventory & Cyclone Performance
Process Data Analytics · CFBC only — **₹1.5 cr** up to, per 100 MW

SEES Analyses bed pressure profile, cyclone DP, loop-seal pressure drop, bed makeup rate and refractory hotspot signatures.

FINDS Detects attrition exceeding makeup, cyclone efficiency loss from refractory wear, and loop-seal blockages disrupting solids return.

ACTS Advises bed makeup, schedules cyclone refractory inspection and warns early of recirculation issues that would otherwise force an outage.

PROBLEMS TODAY Attrition silent until performance degrades · Cyclone refractory wear hidden inside the vessel · Loop-seal blockages cause sudden inventory loss · Bed makeup cost often unaccounted

03 Auxiliary Power Reduction

Run the right auxiliaries, at the right load, at the right time · 4 use cases

13 Aux Pumps + Fans Efficiency Retrace & Load Scheduling
Process Data Analytics · All boilers — **₹6 cr** up to, per 100 MW

SEES Re-traces live efficiency curves for BFPs, ID/FD/PA fans and CW pumps from operating data and compares each against its design curve.

FINDS Detects which units run near design and which have degraded, quantifying the power penalty of running the laggards.

ACTS Advises scheduling to load efficient units first, shift work away from degraded ones and prioritise their maintenance.

PROBLEMS TODAY Rotating aux efficiency degrades silently · Pumps and fans selected by convention, not efficiency · BFP — the largest APC consumer — has an untracked efficiency gap · APC creeps up year on year inside aggregate reporting

14 Intelligent Cooling-Tower Fan Scheduling
Automated Process Control · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses CW supply temperature, ambient wet bulb, load, condenser heat rejection requirement and per-cell fan status and power.

FINDS Detects when fewer cells, or lower VFD speed across more cells, still holds target CW supply temperature.

ACTS Provides cell ON/OFF sequencing that equalises wear, per-cell VFD setpoints and anticipatory adjustment for ambient and load.

PROBLEMS TODAY All cells run regardless of need · Ambient ignored in fan scheduling · Vacuum–power trade-off not optimized · Meaningful power waste in mild weather

15 Mill Parameter Optimization
Automated Process Control · PF + SC + USC — **₹2 cr** up to, per 100 MW

SEES Analyses live coal particle size from computer vision, moisture, grindability inferred from mill power patterns and per-mill LOI feedback.

FINDS Predicts mill power for the current coal source and flags when classifier speed, mill DP, outlet temperature and PA ratio should adapt.

ACTS Coal-adaptive tuning — the same mill run differently per source and blend, valuable where multiple suppliers and washery rejects mix.

PROBLEMS TODAY Mill settings fixed regardless of coal quality · Grindability and moisture variability not adapted to · Operator-dependent classifier settings · Sub-optimal cross-mill bias

16 Optimal ESP TR-Set Setpoint Control
Automated Process Control · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses stack SPM, per-field TR-set power, field current and voltage, rapping schedule and effectiveness, load and dust loading.

FINDS Predicts the collection efficiency needed at current dust loading, and which fields can be trimmed versus held fully energised.

ACTS Closed-loop field-by-field energisation and coordinated rapping, cutting ESP electrical load while holding SPM compliance.

PROBLEMS TODAY ESP fields run at full power regardless of dust load · TR-set power is a significant aux load · Over-generous compliance margins · Manual field tuning rarely done per shift

04 Asset Reliability

Catch degradation weeks before it becomes a trip · 8 use cases

17 Vibration-Based Condition Monitoring
Process Data Analytics · All boilers — **₹5 cr** up to, per 100 MW

SEES Covers every critical rotating asset outside the DCS-monitored turbine train: mills, BFPs, ID/FD/PA fans, CW and CE pumps, feeder and conveyor drives, crushers, stacker-reclaimer, APH and soot-blower drives, ash and limestone handling, FGD blower, compressors, lube oil and hydraulic pumps.

FINDS Analyses vibration signatures (FFT spectrum, time waveform, vector), bearing temperatures, motor current signature and lube oil quality via wired or wireless sensors.

ACTS Detects bearing degradation, misalignment, unbalance, looseness, gearbox wear, lubrication issues and rotor damage **2–8 weeks ahead of failure** — with equipment-tagged alerts, remaining-useful-life estimates and recommended actions.

PROBLEMS TODAY Critical equipment fails without warning · Maintenance reactive on non-Tier-1 equipment · A lost mill, BFP or fan means load reduction or a trip · Aging assets drive insurance and reliability cost

18 Intelligent Soot Blowing Scheduler
Automated Process Control · All boilers — **₹5 cr** up to, per 100 MW

SEES Analyses live Heat-Exchange Index per section, section-specific fouling rate, blower wear history and steam pressure dynamics during blowing.

FINDS Separates sections that genuinely need blowing from time-based blows with negligible heat-exchange impact.

ACTS Pushes closed-loop blow commands to the DCS section by section with post-blow HEI verification — cutting blowing steam and tube erosion, and restoring the SH/RH/economizer absorption profile to reduce downstream attemperation.

PROBLEMS TODAY Time-based blowing wastes steam · Sections cleaned uniformly regardless of fouling · Blowing steam is a meaningful loss · Over-blowing erodes tubes

19 Heat Exchange Index & Section-Level Fouling (incl. APH)
Process Data Analytics · All boilers — **₹3 cr** up to, per 100 MW

SEES Analyses both-side inlet/outlet temperatures and flows for every major section — APH, economizer, primary and final superheater, reheater banks, water walls; for APH also O_2 in/out, hot air, cold gas to ESP and DPs.

FINDS Detects section-specific fouling, scaling, tube plugging and flow short-circuits, and triangulates APH seal leakage across radial, axial, hot- and cold-end seals.

ACTS Gives section-level fouling visibility and seal-specific APH advisory, driving the soot-blowing use case and catching drift that periodic efficiency tests find too late.

PROBLEMS TODAY Fouling degrades efficiency silently · APH seal wear progressive and unseen · Cleaning on calendar, not condition · Cold-end corrosion hidden until damage

20 HP & LP Heater Performance Monitoring
Process Data Analytics · All boilers — **₹3 cr** up to, per 100 MW

SEES Analyses TTD, DCA, feedwater outlet temperature, drain flows, extraction pressures and bypass valve positions across HP heaters, deaerator and LP heaters.

FINDS Detects per-heater degradation from fouling, tube damage, drain malfunction, air ingress and bypass leakage.

ACTS Heater-by-heater advisory with feedwater terminal temperature recovery, restoring efficiency on a cycle that drifts silently.

PROBLEMS TODAY Heater degradation is silent · Bypass valves open unnoticed · Sub-optimal drain handling · TTD/DCA rarely tracked

21 ESP Health Monitoring
Process Data Analytics · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses per-field TR-set current and voltage patterns, insulator leakage current, rapping effectiveness, hopper levels and SPM excursion correlation.

FINDS Detects insulator failures before TR-set tripping, plate ash build-up, rapping failures, hopper bridging and TR-set component degradation.

ACTS Replaces time-based ESP overhauls with condition-based scheduling and location-tagged alerts before issues become compliance violations.

PROBLEMS TODAY ESP failures cause SPM spikes · Field-level issues hidden until breakdown · Insulator failures common but unpredictable · Time-based maintenance

22 Generator Winding + Air Cooler Health
Process Data Analytics · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses stator and rotor winding temperatures, cooler inlet/outlet temperatures, cooling water flow, partial discharge sensors where fitted and leakage flux.

FINDS Detects winding hot spots from restricted cooling, seasonal cooler effectiveness loss, and partial discharge as an insulation-degradation precursor.

ACTS Provides a generator health view, insulation aging-rate tracking and cooler maintenance advisory.

PROBLEMS TODAY Generator failures rare but catastrophic · Winding deterioration silent · Cooler effectiveness untracked across seasons · PD measured only on instrumented units

23 Corrosion & Scaling Surveillance
Process Data Analytics · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses feedwater pH, conductivity and dissolved oxygen, drum water chemistry, dosing residuals, blowdown and the historic tube-failure record.

FINDS Detects chemistry excursions in real time and predicts tube corrosion, scaling and turbine deposit risk from exposure history.

ACTS Chemistry-to-tube-life and dosing advisory — the long-term integrity management that spot checks miss.

PROBLEMS TODAY Chemistry excursions cause tube damage years later · Chemistry–failure linkage is tribal knowledge · Monitoring not continuous · Corrections only after failure

24 Condensate Contamination & Condenser Tube Leak Detection
Process Data Analytics · All boilers — **₹2 cr** up to, per 100 MW

SEES Analyses condensate conductivity, sodium and silica trends, hardness and boiler makeup water consumption.

FINDS Detects condenser tube leaks from the condensate chemistry signature — typically the fastest reliable method.

ACTS Early warning, localisation to the leaking water box and tube-isolation advisory, before CW chemistry silently damages boiler tubes.

PROBLEMS TODAY Tube leaks contaminate condensate · Resulting boiler tube damage is slow and costly to unwind · Manual rounds inadequate for early detection · Water-box localisation is manual

NEXT STEP

Start with a plant visit

Share a sample of DCS, lab and design data and we will come back with a shortlist of use cases sized against your plant's own loss profile.

www.faclonlabs.com